Shantanu Dhotkar

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Education

Vishwakarma Institute of Information Technology (VIIT), Pune 2023 – 2027
B.Tech. in Electronics and Telecommunication Engineering *CGPA: 8.48 / 10.0*

Technical Skills

Programming	Embedded C (Bare Metal Programming), STM32 Hal-programming, C++, Arduino Programming, Python (Basic), 8051 Assembly
MCUs & SoCs	NUCLEO-F446RE, STM32f401CCU6, STM32F103C8T6, ESP32, ESP8266, Arduino Uno/Nano, AT89S52 (8051)
Architectures	ARM Cortex-M3, ARM Cortex-M4, ARM7-LPC2148, 8051 (CISC)
Protocols	UART/USART, SPI, I2C, RS485, Modbus RTU, MQTT, CAN Bus
RTOS	FreeRTOS (Tasks, Queues, Mutex, Semaphores, Event Groups), Bare-metal Porting (Cortex-M4)
Hardware Design	PCB Design (KiCad), Soldering (THT & SMD), Circuit Debugging, Signal Measurement
Tools & IDEs	STM32CubeIDE, VS Code (PlatformIO), Proteus, Multisim, Git and GitHub, MATLAB Simulink, SWD-Debugging

Projects

Remote Terminal Unit | *STM32F401CCU6, ESP8266, RS485, Modbus RTU, MQTT* *Dec 2025 – Mar 2026*

- Architected an industrial IoT gateway on STM32F401CCU6 acquiring 16 DIO (MCP23S17 SPI), 4 AIO (4–20 mA, 12-bit ADC, ±**0.1%**), and energy parameters from Schneider EM6436H via RS485 Modbus RTU (8E1, CRC-16) at 1-second cycle.
- Developed embedded firmware in STM32 HAL covering RS485 (echo-discard, TC-flag half-duplex, ORE/PE/FE recovery), SPI, ADC-DMA, and custom raw MQTT library (CONNECT/PUBLISH/PINGREQ); deployed ESP8266 (AT firmware) to publish JSON telemetry to HiveMQ.

Smart Room Monitor with FreeRTOS | *STM32F401CCU6, FreeRTOS, DHT11, HC-SR04, LCD, UART* *Jan 2026 – Feb 2026*

- Ported FreeRTOS v202406 bare-metal from scratch on STM32F401CCU6 without CubeMX, manually configuring Cortex-M4 SysTick, PendSV, and SVCall handlers and authoring `FreeRTOSConfig.h` for 84 MHz clock, heap_4 memory model, and stack-overflow detection.
- Built a 4-task concurrent monitor reading DHT11 and HC-SR04 via DWT cycle-counter timed bit-bang drivers; shared sensor data through `SensorMsg_t` queue (`xQueueOverwrite/xQueuePeek`), protected shared I2C and UART buses with dual Mutexes, and triggered LCD alerts via a 3-bit Event Group on temperature > 35 °C, humidity > 80%, and distance < 20 cm threshold breaches.

Pzem004t Energy Monitor with RS485 Protocol | *ESP32, RS485, Modbus RTU, PZEM-004T, MAX485* *Aug 2025 – Oct 2025*

- Acquired V, I, P, energy, frequency, and PF from PZEM-004T via Modbus RTU over RS485; implemented MAX485 UART-to-RS485 conversion at both ends for half-duplex communication on ESP32 (±**0.5%** accuracy, 15-second cycle).
- Deployed Wi-Fi telemetry on ESP32 with Telegram Bot API for real-time remote monitoring and threshold-based alert notifications.

Experience

Team Vishwaracers BAJA – SAE India BAJA Feb 2024 – Jan 2025
Electrical Member | All-Terrain Vehicle (ATV) Racing Team *Pune, India*

- Developed tractive system and low-voltage electrical architecture for a competitive BAJA ATV; contributed to a Data Acquisition System (DAS) for real-time vehicle telementry, worked with CAN bus automotive Protocol.
- Gained hands-on exposure to BMS, PMSM motors, and motor controllers under real-world safety standards and competition timelines.

Indian Air Force – 9 Base Repair Depot (9BRD) Jun 2025 – Jul 2025
Technical Training Intern *Pune,India*

- Completed 6-week technical training at a 9 Base Repair Depot; gained exposure to Electronic Counter Measure (ECM), PCB Testing and Repairing, and precision equipment handling.

Research, Certifications & Achievements

- **Research Paper:** Smart Cargo Monitoring System | *Grenze International Journal* (2024) | Indexed in Scopus
- **Certification:** Microcontroller Embedded C Programming | *Udemy* (2025–2026)
- **Top 7 Teams** – College-Level Hackathon, VIIT Pune (Mar 2025)
- **Participant** – **Kurukshetra 2025** National Level 24-Hour Hackathon (Aug 2025)